

Q.1 Given

①

$$E \rightarrow E + T \mid T$$

$$T \rightarrow T \times F \mid F$$

$$F \rightarrow (E) \mid a$$

parse trees and derivations.

a. a

$$E \rightarrow T$$

$$T \rightarrow F$$

$$F \rightarrow a$$



b. a+a

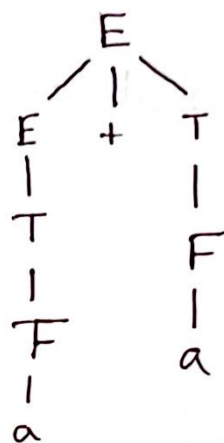
$$E \rightarrow E + T$$

$$E \rightarrow T + T$$

$$T \rightarrow F + T$$

$$F \rightarrow a + T$$

$$F \rightarrow a$$



c. a+a+a

$$E \rightarrow E + T$$

$$E \rightarrow E + T + T$$

$$E \rightarrow T + T + T$$

$$T \rightarrow F + T + T$$

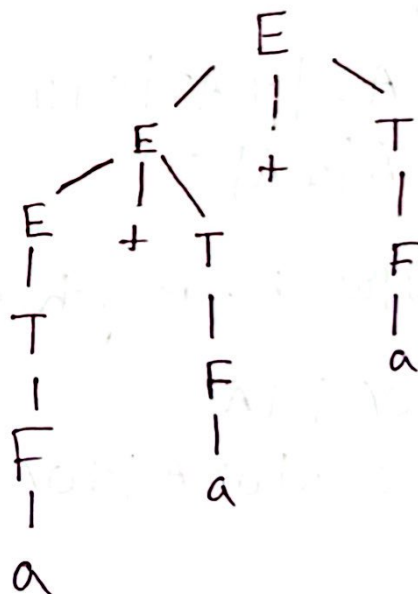
$$F \rightarrow a + T + T$$

$$T \rightarrow F + T$$

$$F \rightarrow a + T$$

$$T \rightarrow F$$

$$F \rightarrow a$$



d. ((a))

$E \rightarrow T$

$T \rightarrow F$

$F \rightarrow (E)$

$E \rightarrow T$

$T \rightarrow F$

$F \rightarrow (E)$

$E \rightarrow T$

$T \rightarrow F$

$F \rightarrow a$

E
 $|$
 T
 $|$
 F
 $|$
 (E)
 $|$
 T
 $|$
 F
 $|$
 (E)
 $|$
 T
 $|$
 F
 $|$
 a

2.4

a. $L = \{w \mid w \text{ contains at least three } 1s\}$

$S \rightarrow A1A1A1B$

$A \rightarrow 0A \mid 1A \mid \epsilon$

$B \rightarrow 0B \mid 1B \mid \epsilon$

b. $L = \{w \mid w \text{ starts and ends with same symbol}\}$

$S \rightarrow 0A0 \mid 1A1 \mid 0 \mid 1$

$A \rightarrow 0A \mid 1A \mid \epsilon$

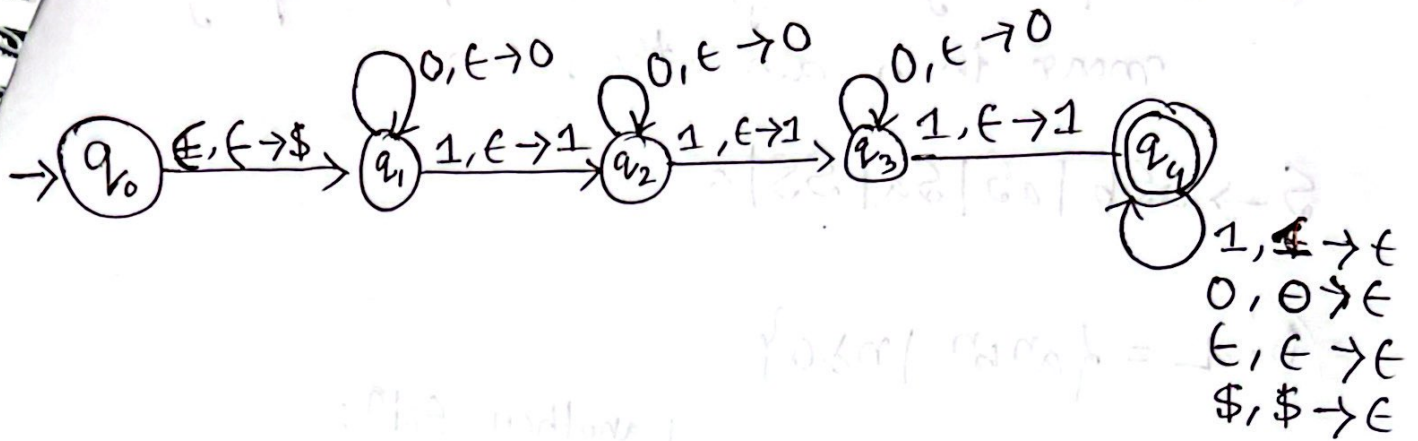
c. $L = \{w \mid \text{the length of } w \text{ is odd}\}$

$S \rightarrow 0A \mid 1A$

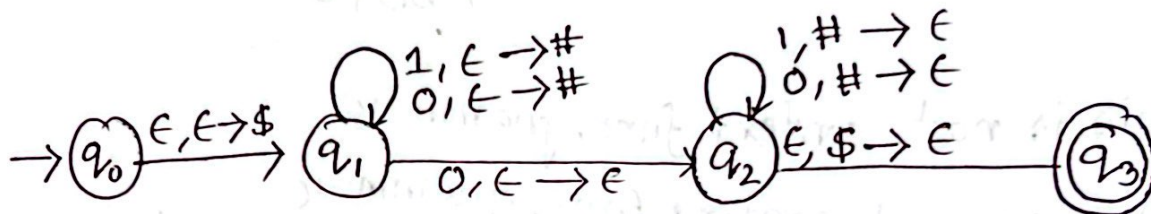
$A \rightarrow 00A \mid 01A \mid 10A \mid 11A \mid \epsilon$

(3)

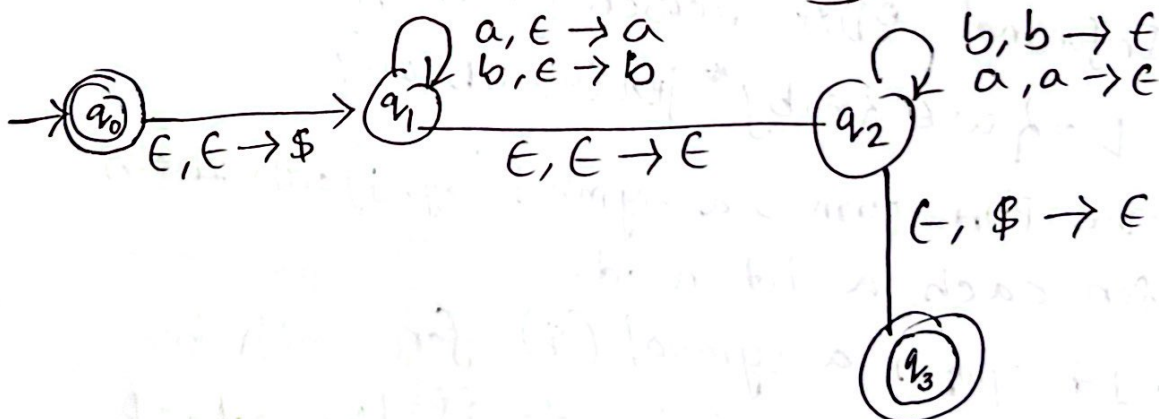
a. $L = \{w \mid w \text{ contains at least three } 1\}$



d. $L = \{w \mid \text{the length of } w \text{ is odd and middle symbol is } a \text{ or } 0\}$



e. $L = \{ww^R \mid w \text{ is palindrome}\}$



2.6

- a. $L = \{ \text{set of strings over the alphabet } \{a, b\} \text{ with more than a's than b's} \}$

$$S \rightarrow aSb \mid aS \mid Sa \mid SS \mid a$$

b. $L = \{ a^n b^n \mid n \geq 0 \}$

$$S \rightarrow aS \mid bS \mid aSb \mid T$$

$$T \rightarrow aT \mid bT \mid abTa \mid \epsilon$$

another solⁿ:

$$S \rightarrow aSb \mid A \mid B \mid AB \mid BA$$

$$A \rightarrow aA \mid a$$

$$B \rightarrow bB \mid b$$

- c. This is not context free grammar

- d. This is not context free grammar

2.7

~~b. $L = \{ w \in \{a, b\}^* \mid w \notin \{a^n\} \}$~~

- a. Informal PDA Description of the

$$L = \{ w \in \{a, b\}^* \mid \#a(w) \neq \#b(w) \}$$

- The PDA pushes a symbol (say, \$) onto stack for each a it reads.
- It pops a symbol (\$) for each b.
- At the end of input if the stack contains at least one remaining symbol it accepts.
- This ensures that the number of a's is strictly greater

• Informal description for the Language

$$L = \{a^n b^n \mid n \geq 0\}$$

- In an empty stack \$
- Then if we read 'a' push 'a' in the stack until we don't get 'b'
- For every 'b' we have to pop 'a' from the stack.
- If the entire input is consumed and the top of the stack is '\$' we pop it and accept the PDA.

2.11 Given, CFG,

$$\begin{aligned} E &\rightarrow E + T \mid T \\ T &\rightarrow T \times F \mid F \\ F &\rightarrow (E) \mid a \end{aligned}$$

Remove Unit production.

Step 1.

$$\begin{aligned} E_0 &\rightarrow E \\ E &\rightarrow E + T \mid T \\ T &\rightarrow T \times F \mid F \\ F &\rightarrow (E) \mid a \end{aligned}$$

Remove Unit Production, $E_0 \rightarrow E, E \rightarrow T, T \rightarrow F$

$$\begin{aligned} E_0 &\rightarrow E + T \mid T \times F \mid (E) \mid a \\ E &\rightarrow E + T \mid T \times F \mid (E) \mid a \end{aligned}$$

$$T \rightarrow TXF \mid (E) \mid a$$

$$F \rightarrow (E) \mid a$$

Step - 3:

$$E_0 \rightarrow \cancel{EA \mid TB \mid (E) \mid a} \quad EA \mid TB \mid (E) \mid a$$

$$E \rightarrow EA \mid TB \mid (E) \mid a$$

$$T \rightarrow TB \mid (E) \mid a$$

$$F \rightarrow (E) \mid a$$

$$A \rightarrow +T$$

$$B \rightarrow \times F$$

~~$\epsilon \rightarrow$~~

Now Transition for the PDA

$$\delta(q_0, \epsilon, E_0) \rightarrow (q_0, \cancel{EA})$$

$$\delta(q_0, \epsilon, E_0) \rightarrow (q_0, TB) \quad \delta(q_0, \epsilon, E_0) \rightarrow (q_0, TB)$$

$$\delta(q_0, \epsilon, E_0) \rightarrow (q_0, (E)) \quad \delta(q_0, \epsilon, E_0) \rightarrow (q_0, (E))$$

$$\delta(q_0, \epsilon, E_0) \rightarrow (q_0, a) \quad \delta(q_0, \epsilon, E_0) \rightarrow (q_0, a)$$

$$\delta(q_0, \epsilon, E) \rightarrow (q_0, EA)$$

$$\delta(q_0, \epsilon, E) \rightarrow (q_0, TB)$$

$$\delta(q_0, \epsilon, E) \rightarrow (q_0, (E))$$

$$\delta(q_0, \epsilon, E) \rightarrow (q_0, a)$$

$$\delta(q_0, \epsilon, T) \rightarrow (q_0, TB)$$

$$\delta(q_0, \epsilon, T) \rightarrow (q_0, (E))$$

$$\delta(q_0, \epsilon, T) \rightarrow (q_0, a)$$

$$\delta(q_0, \epsilon, F) \rightarrow (q_0, (E))$$

$$\delta(q_0, \epsilon, F) \rightarrow (q_0, a)$$

$$\begin{aligned}
 \delta(q_0, \epsilon, A) &\rightarrow (q_0, +T) \\
 \delta(q_0, \epsilon, B) &\rightarrow (q_0, XF) \\
 \delta(q_0, a, a) &\rightarrow (q_0, \epsilon) \\
 \delta(q_0, U, U) &\rightarrow (q_0, \epsilon) \\
 \delta(q_0, X, X) &\rightarrow (q_0, \epsilon)
 \end{aligned}$$

This are the transition function for PDA.

2.14

Given, CFG

$$A \rightarrow BAB | B | \epsilon$$

$$B \rightarrow 00 | \epsilon$$

~~Remove Null~~

Step 1:

$$A_0 \rightarrow A$$

$$A \rightarrow BAB | B | \epsilon$$

$$B \rightarrow 00 | \epsilon$$

Step 2

Remove Null production.

$$A_0 \rightarrow \cancel{BAB} | \cancel{AB} | \cancel{BA} | A | B | A$$

$$A \rightarrow \cancel{BAB} | \cancel{AB} | \cancel{BA} | A | B$$

$$B \rightarrow 00$$

Step 3

Remove Unit Production.

$$A_0 \rightarrow \cancel{A} \cdot BAB | AB | BA$$

$$A \rightarrow \cancel{A} \cdot BAB | AB | BA$$

$$B \rightarrow 00$$

Step: 4

$A_0 \rightarrow Bx \mid AB \mid BA$

$A \rightarrow Bx \mid AB \mid BA$

$x \rightarrow AB$

$B \rightarrow 0$

This is the CNF of the given CFG

3.2

(a) 11

Input tape

1 1 $\sqcup$

Configuration

• q_1 1 1

• x q_3 1

• x 1 q_3 — accept.

(b) 10#11

Input tape 1 0 # 1 1 $\sqcup$

Configuration,

q_1 1 0 # 11

x q_3 0 # 11

x 0 q_3 # 11

x 0 # q_5 11

x 0 # x q_6 1

x 0 # q_6 # x 1

x 0 q_6 # x 1

x q_6 0 # x 1

q_6 x 0 # x 1

q_1 x 0 # x 1

$xq_1 0 \# x 1$

$x x q_2 \# x 1$

$x x q_2 \# x 1$

$x x \# q_4 x 1$

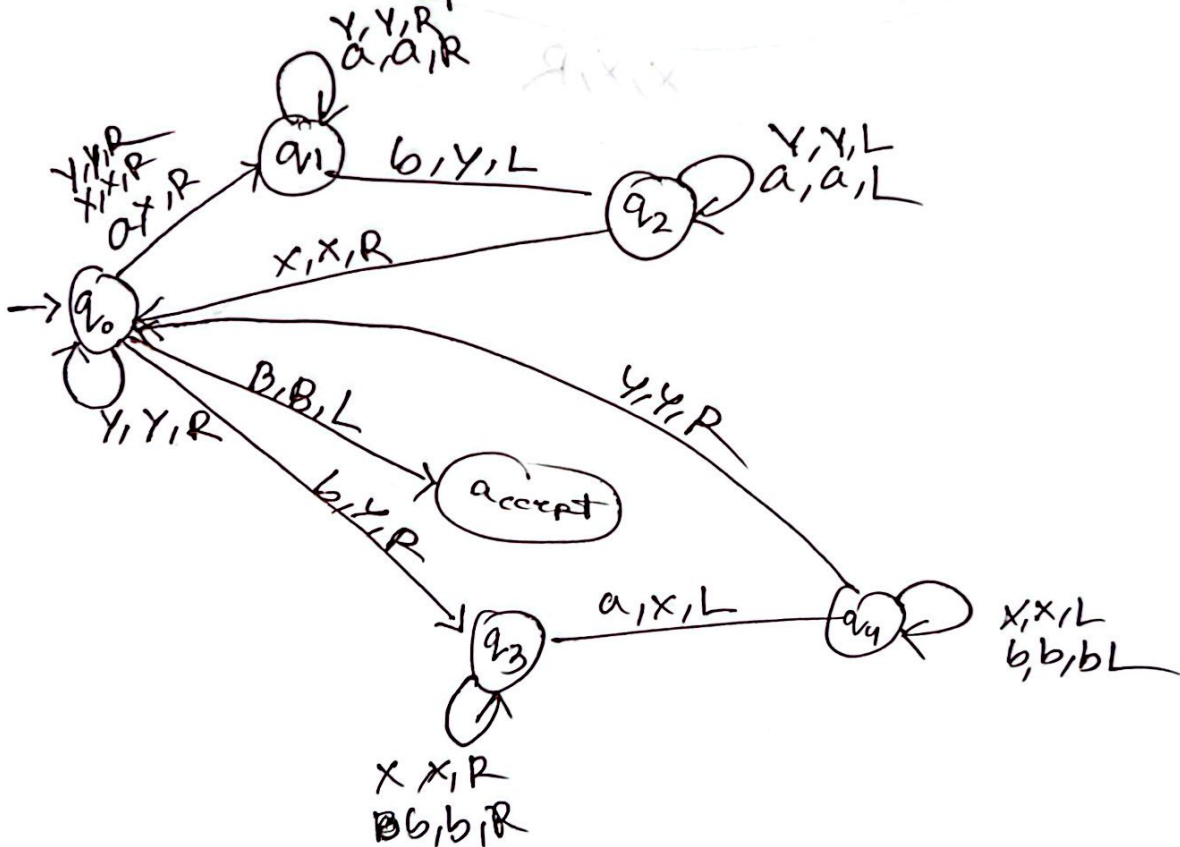
$x x \# x q_4 1$

$x x \# x 1 q_4$

- Reject.

3.8

a. $L = \{ w \mid w \text{ contains equal } 0\text{'s and } 1\text{'s} \}$



3.8. b

$$L = \{w \mid w = a^n b^{2n} \mid n \geq 1\}$$

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